TIME SERIES ANALYSIS - STOCK PRICE DATASET

In [1]:
```
import pandas as pd

sp = pd.read_csv("2) Stock Prices Data Set.csv")
```

In [2]:
```
sp.head()
```

Out[2]:

	symbol	date	open	high	low	close	volume
0	AAL	2014-01-02	25.0700	25.8200	25.0600	25.3600	8998943
1	AAPL	2014-01-02	79.3828	79.5756	78.8601	79.0185	58791957
2	AAP	2014-01-02	110.3600	111.8800	109.2900	109.7400	542711
3	ABBV	2014-01-02	52.1200	52.3300	51.5200	51.9800	4569061
4	ABC	2014-01-02	70.1100	70.2300	69.4800	69.8900	1148391

In [3]:
```
sp.describe()
```

Out[3]:

	open	high	low	close	volume
count	497461.000000	497464.000000	497464.000000	497472.000000	4.974720e+05
mean	86.352275	87.132562	85.552467	86.369082	4.253611e+06
std	101.471228	102.312062	100.570957	101.472407	8.232139e+06
min	1.620000	1.690000	1.500000	1.590000	0.000000e+00
25%	41.690000	42.090000	41.280000	41.703750	1.080166e+06
50%	64.970000	65.560000	64.353700	64.980000	2.084896e+06
75%	98.410000	99.230000	97.580000	98.420000	4.271928e+06
max	2044.000000	2067.990000	2035.110000	2049.000000	6.182376e+08

In [4]:
```
sp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 497472 entries, 0 to 497471
Data columns (total 7 columns):
 #   Column  Non-Null Count   Dtype  
---  ------  --------------   -----  
 0   symbol  497472 non-null  object 
 1   date    497472 non-null  object 
 2   open    497461 non-null  float64
 3   high    497464 non-null  float64
 4   low     497464 non-null  float64
 5   close   497472 non-null  float64
 6   volume  497472 non-null  int64  
dtypes: float64(4), int64(1), object(2)
memory usage: 26.6+ MB
```

In [6]:
```
sp["symbol"].nunique()
```

Out[6]:
```
505
```

In [7]:
```
sp["symbol"].value_counts()
```

Out[7]:
```
symbol
AAL     1007
PRGO    1007
NVDA    1007
NUE     1007
NTRS    1007
        ... 
DXC      189
BHGE     126
BHF      117
DWDP      83
APTV      18
Name: count, Length: 505, dtype: int64
```

In [8]:
```
sp.isnull().sum()
```

Out[8]:
```
symbol     0
date       0
open      11
high       8
low        8
close      0
volume     0
dtype: int64
```

In [9]:
```
# Identify most occuring symbols with complete entries

symbol_count = sp["symbol"].value_counts()
max_symbol_count = symbol_count.max()
top_symbol = symbol_count[symbol_count == max_symbol_count].index[:5]

top_symbol
```

Out[9]:
```
Index(['AAL', 'PRGO', 'NVDA', 'NUE', 'NTRS'], dtype='object', name='symbol')
```

In [10]:
```
top5symbols = sp[sp["symbol"].isin(top_symbol)].copy()
top5symbols
```

Out[10]:

	symbol	date	open	high	low	close	volume
0	AAL	2014-01-02	25.07	25.820	25.060	25.36	8998943
328	NTRS	2014-01-02	61.76	61.950	60.665	60.89	1011050
329	NUE	2014-01-02	53.29	53.380	52.510	52.73	1611326
330	NVDA	2014-01-02	15.92	15.980	15.720	15.86	6502296
363	PRGO	2014-01-02	152.45	154.520	152.110	153.10	1209456
...	...	...	...	...	...	...	...
496967	AAL	2017-12-29	52.42	52.820	52.010	52.03	2697096
497311	NTRS	2017-12-29	100.56	101.455	99.860	99.89	637509
497312	NUE	2017-12-29	64.75	64.750	63.570	63.58	1519439
497313	NVDA	2017-12-29	198.46	198.460	193.500	193.50	6999116
497346	PRGO	2017-12-29	87.57	87.790	87.040	87.16	442315

5035 rows × 7 columns

In [11]:

```
top5symbols["date"] = pd.to_datetime(top5symbols["date"])
```

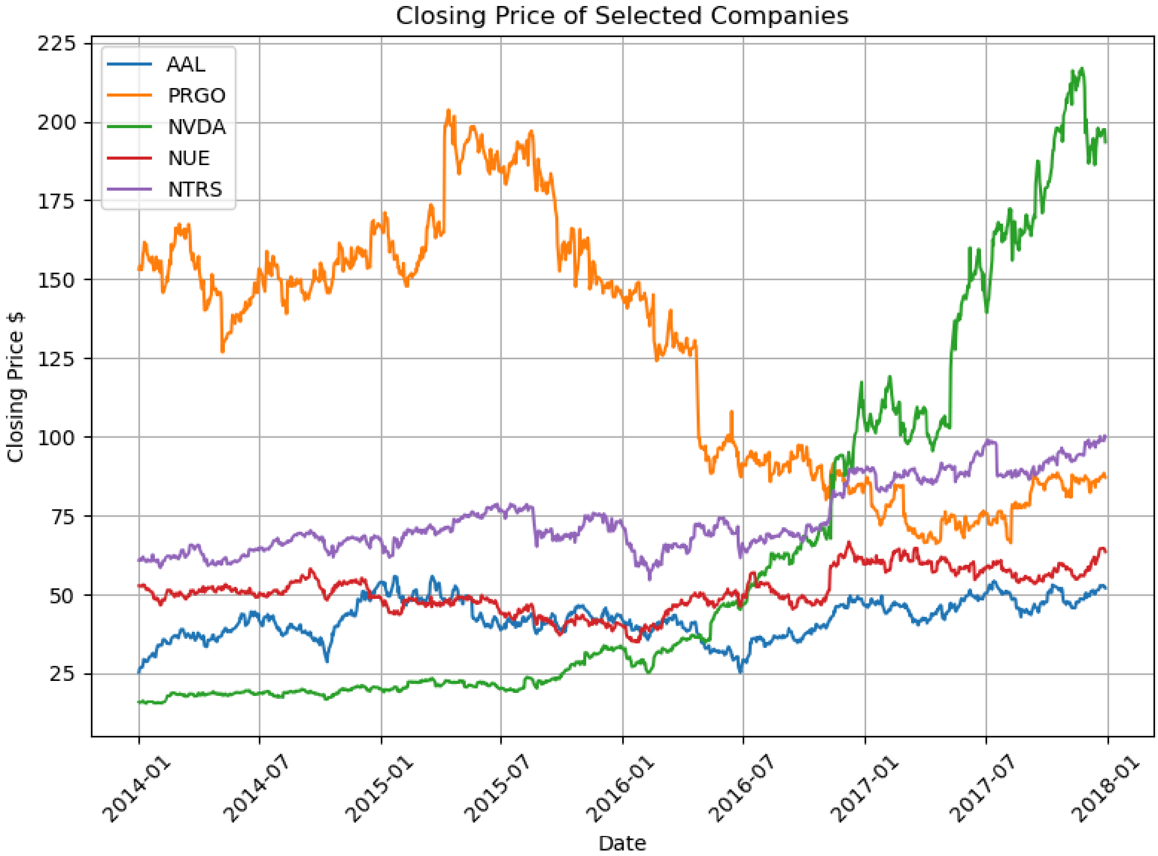
In [12]:

```
import matplotlib.pyplot as plt
import seaborn as sb
```

EXPLORATORY TIME ANALYSIS- CHANGES IN CLOSING PRICE OF SELECT STOCK OVER TIME

In [13]:

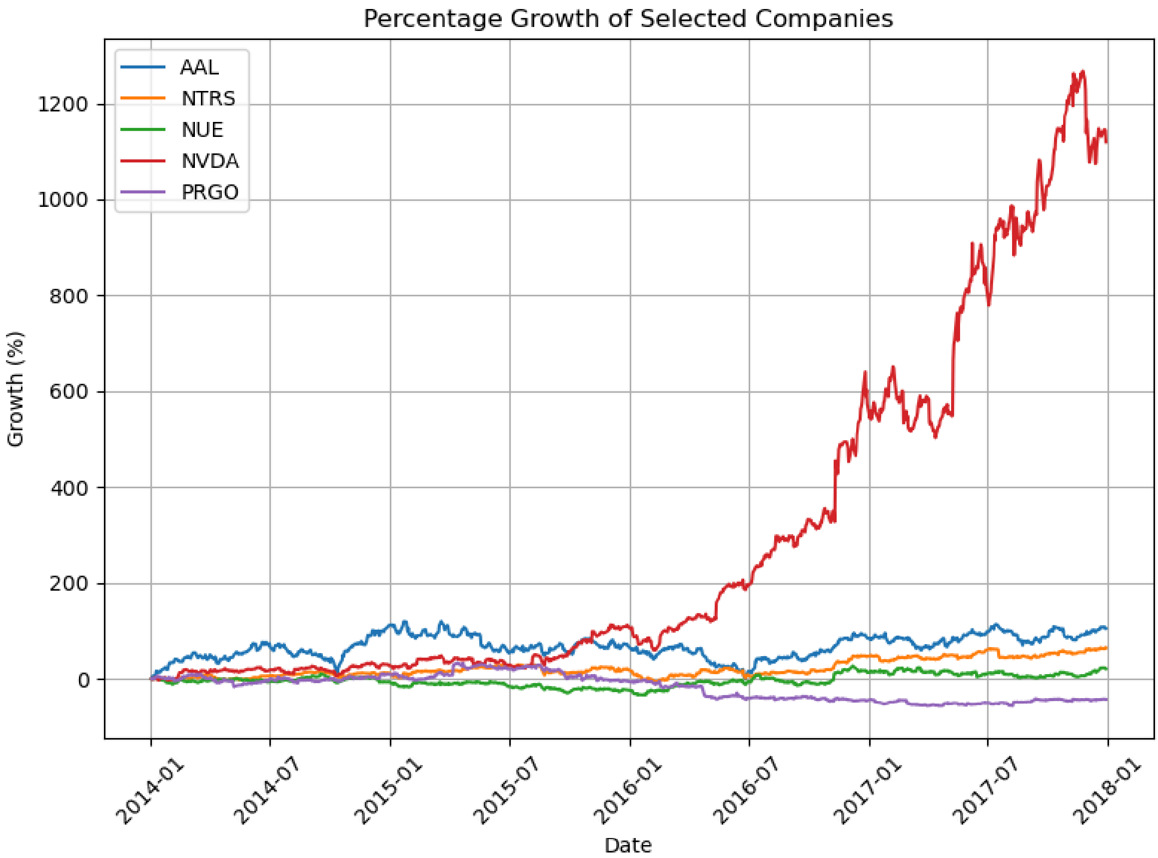
```
plt.figure(figsize=(8,6))

for symbol in top_symbol:
    company = top5symbols[top5symbols["symbol"]==symbol]
    plt.plot(company["date"], company["close"], label=symbol)

plt.title("Closing Price of Selected Companies")
plt.xlabel("Date")
plt.ylabel("Closing Price $")
plt.grid(True)
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```



PERCENTAGE GROWTH ANALYSIS

In [14]:
```
symbols = top5symbols["symbol"].unique()
```

In [15]:
```
plt.figure(figsize=(8, 6))

for symbol in symbols:
    company = top5symbols[top5symbols["symbol"] == symbol].copy()

    company["pct_growth"] = (company["close"] / company["close"].iloc[0] - 1) *

    plt.plot(company["date"], company["pct_growth"], label=symbol)

plt.title("Percentage Growth of Selected Companies")
plt.xlabel("Date")
plt.ylabel("Growth (%)")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```



Key Insight – Percentage Growth

Percentage Growth compares today's price with the first day's price. Among the selected companies, NVDA recorded the strongest cumulative growth over the given period, significantly outperforming its peers. In contrast, PRGO experienced the weakest growth even incurring losses, making these two companies suitable candidates for deeper time-series analysis.

DAILY VOLATILITY

In [16]:
```
symbols = top5symbols["symbol"].unique()

volatility_summary = []

for symbol in symbols:
    company = top5symbols[top5symbols["symbol"]==symbol].copy()
    company = company.sort_values("date")

    daily_return = company["close"].pct_change() * 100

    company["daily_return"] = daily_return

    volatility = company["daily_return"].std()

    volatility_summary.append({"Symbol":symbol, "Volatility":volatility})
    print(volatility_summary)
```

```
[{'Symbol': 'AAL', 'Volatility': 2.238868341352129}]
[{'Symbol': 'AAL', 'Volatility': 2.238868341352129}, {'Symbol': 'NTRS', 'Volatili
ty': 1.3806060722295004}]
[{'Symbol': 'AAL', 'Volatility': 2.238868341352129}, {'Symbol': 'NTRS', 'Volatili
ty': 1.3806060722295004}, {'Symbol': 'NUE', 'Volatility': 1.6674995186555477}]
[{'Symbol': 'AAL', 'Volatility': 2.238868341352129}, {'Symbol': 'NTRS', 'Volatili
ty': 1.3806060722295004}, {'Symbol': 'NUE', 'Volatility': 1.6674995186555477},
{'Symbol': 'NVDA', 'Volatility': 2.3738853039732764}]
[{'Symbol': 'AAL', 'Volatility': 2.238868341352129}, {'Symbol': 'NTRS', 'Volatili
ty': 1.3806060722295004}, {'Symbol': 'NUE', 'Volatility': 1.6674995186555477},
{'Symbol': 'NVDA', 'Volatility': 2.3738853039732764}, {'Symbol': 'PRGO', 'Volatil
ity': 2.142584660947524}]
```

In [17]:
```
volatility_summary =pd.DataFrame(volatility_summary)
volatility_summary
```

Out[17]:

	Symbol	Volatility
0	AAL	2.238868
1	NTRS	1.380606
2	NUE	1.667500
3	NVDA	2.373885
4	PRGO	2.142585

In [18]:
```
volatility_summary["Volatility"] = volatility_summary["Volatility"].round(2)
volatility_summary
```

Out[18]:

	Symbol	Volatility
0	AAL	2.24
1	NTRS	1.38
2	NUE	1.67
3	NVDA	2.37
4	PRGO	2.14

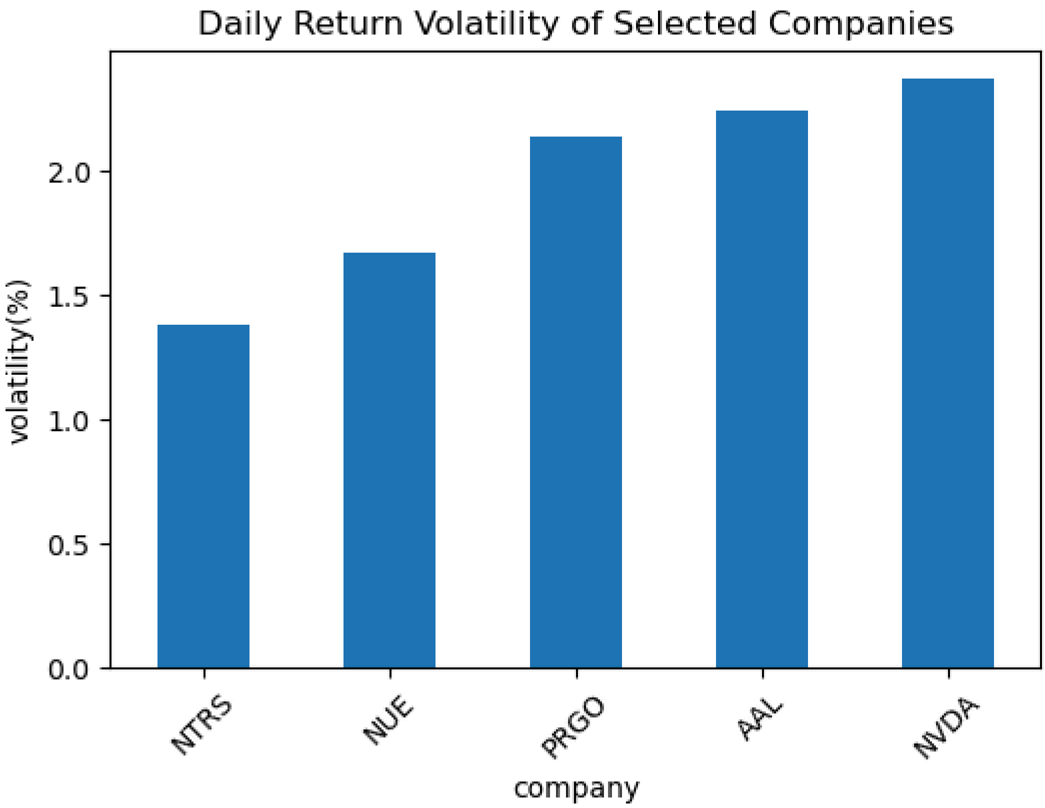
In [19]:
```
volatility_ranked = volatility_summary.sort_values("Volatility")
volatility_ranked
```

Out[19]:

	Symbol	Volatility
1	NTRS	1.38
2	NUE	1.67
4	PRGO	2.14
0	AAL	2.24
3	NVDA	2.37

In [20]:
```
volatility_ranked.index = volatility_ranked["Symbol"]
```

```
volatility_ranked["Volatility"].plot(kind="bar", figsize=(6,4))
plt.title("Daily Return Volatility of Selected Companies")
plt.xlabel("company")
plt.ylabel("volatility(%)")
plt.xticks(rotation=45)

plt.show()
```



Key Insight – Volatility

NVDA with the highest volatility of 2.37% shows that it has the **largest daily price fluctuations** among the selected companies during the period covered. This is interesting because, earlier I found that **NVDA also had the highest cumulative growth**. That suggests an important investment principle: **Higher returns often come with higher risk**.

NTRS with the lowest volatility of 1.38% suggests a **relatively stable price fluctuation**, making it the most stable amongst the five. Eventhough it had a relatively low percentage growth.

PRGO showed a **relatively high volatility** at 2.14% and the **weakest growth performance** with losses incurred. It goes without saying that investors experienced relatively large daily fluctuations without enjoying strong long-term growth; this is not encouraging

These findings indicate the risk and return involved in stock trading.

NVDA and **PRGO** will be subjects for further analysis due to their peculiarity as best behaving and worst behaving in cummulative growth

SEASONAL DECOMPOSITION

1. DECOMPOSITION OF NVDA

In [21]:
```
nvda = top5symbols[top5symbols["symbol"]=="NVDA"].copy()
nvda
```

Out[21]:

	symbol	date	open	high	low	close	volume
330	NVDA	2014-01-02	15.92	15.9800	15.7200	15.86	6502296
813	NVDA	2014-01-03	15.89	15.9201	15.6200	15.67	6483288
1296	NVDA	2014-01-06	15.83	16.0000	15.6800	15.88	10237417
1779	NVDA	2014-01-07	16.04	16.2000	15.9250	16.14	8332170
2262	NVDA	2014-01-08	16.20	16.4400	16.1400	16.36	7704980
...	...	...	...	...	...	...	...
495293	NVDA	2017-12-22	194.40	195.6500	191.2500	195.27	11656871
495798	NVDA	2017-12-26	193.03	197.7500	191.8200	197.44	8877033
496303	NVDA	2017-12-27	196.90	199.9700	196.3100	197.17	8237223
496808	NVDA	2017-12-28	198.13	199.3800	197.1514	197.40	6006248
497313	NVDA	2017-12-29	198.46	198.4600	193.5000	193.50	6999116

1007 rows × 7 columns

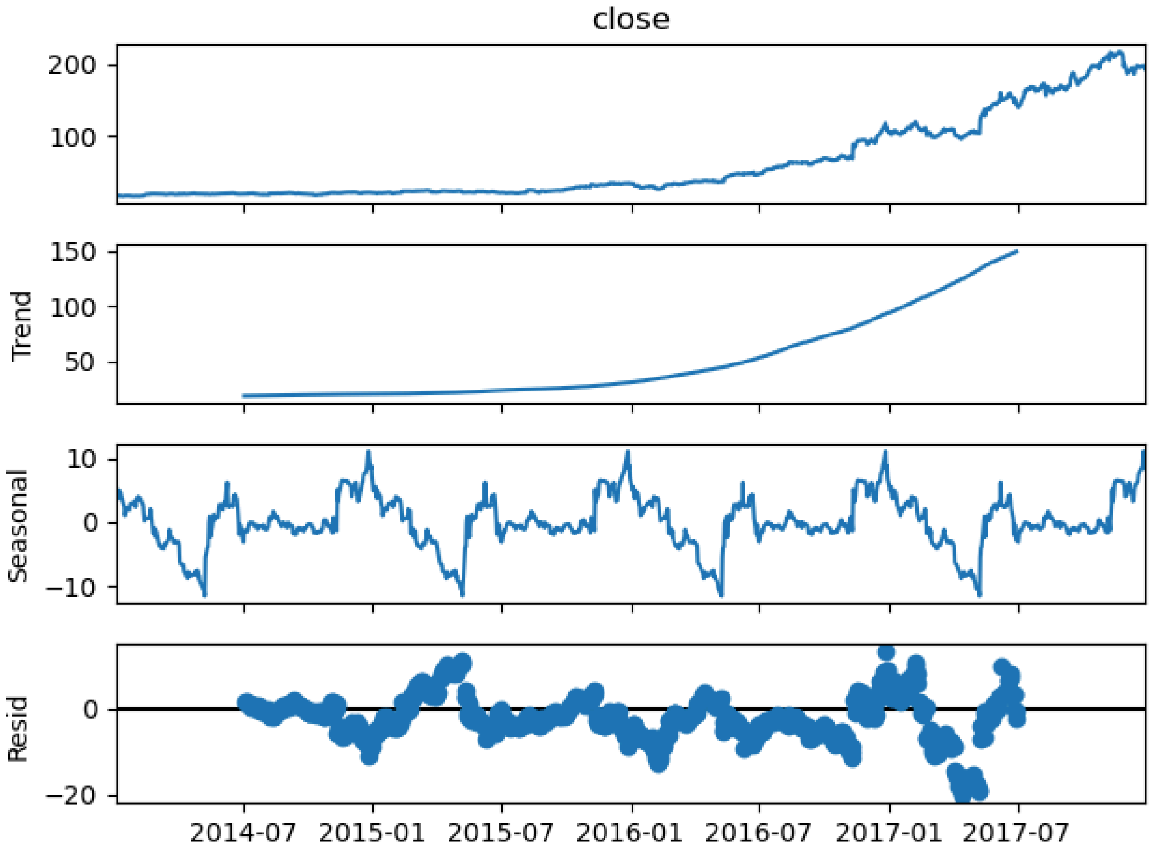
In [22]:
```
nvda["date"] = pd.to_datetime(nvda["date"])

nvda = nvda.sort_values("date")

nvda = nvda.set_index("date")

nvda.head()
```

Out[22]:

	symbol	open	high	low	close	volume
date						
2014-01-02	NVDA	15.92	15.9800	15.720	15.86	6502296
2014-01-03	NVDA	15.89	15.9201	15.620	15.67	6483288
2014-01-06	NVDA	15.83	16.0000	15.680	15.88	10237417
2014-01-07	NVDA	16.04	16.2000	15.925	16.14	8332170
2014-01-08	NVDA	16.20	16.4400	16.140	16.36	7704980

In [23]:
```
from statsmodels.tsa.seasonal import seasonal_decompose
```

In [24]:
```
decomposition = seasonal_decompose(
    nvda["close"],
    model="additive",
    period=252)

decomposition.plot()

plt.show()
```



The **decomposition of NVDA's daily closing prices** reveals a slow increase in closing price during the start period. Over time, it experienced a **continuous upward trend**, particularly from 2016 onward, although having some fluctuations it sustained an **accelerating growth** over the study period. The **seasonal component** exhibits recurring fluctuations based on the assumed annual trading cycle (252 trading days), but these fluctuations are relatively small compared with the overall price movement, suggesting that seasonality is not the primary driver of the stock's performance. The **residual component** is centered around zero with several notable spikes, indicating irregular price movements likely associated with broader market events. **Overall**, the analysis suggests that NVDA's stock performance during the study period was driven predominantly by long-term growth rather than seasonal effects, while still being influenced by unpredictable market volatility.

2. DECOMPOSITION OF PRGO

In [25]:
```
prgo = top5symbols[top5symbols["symbol"]=="PRGO"].copy()
prgo["date"] = pd.to_datetime(prgo["date"])

prgo = prgo.sort_values("date")
```

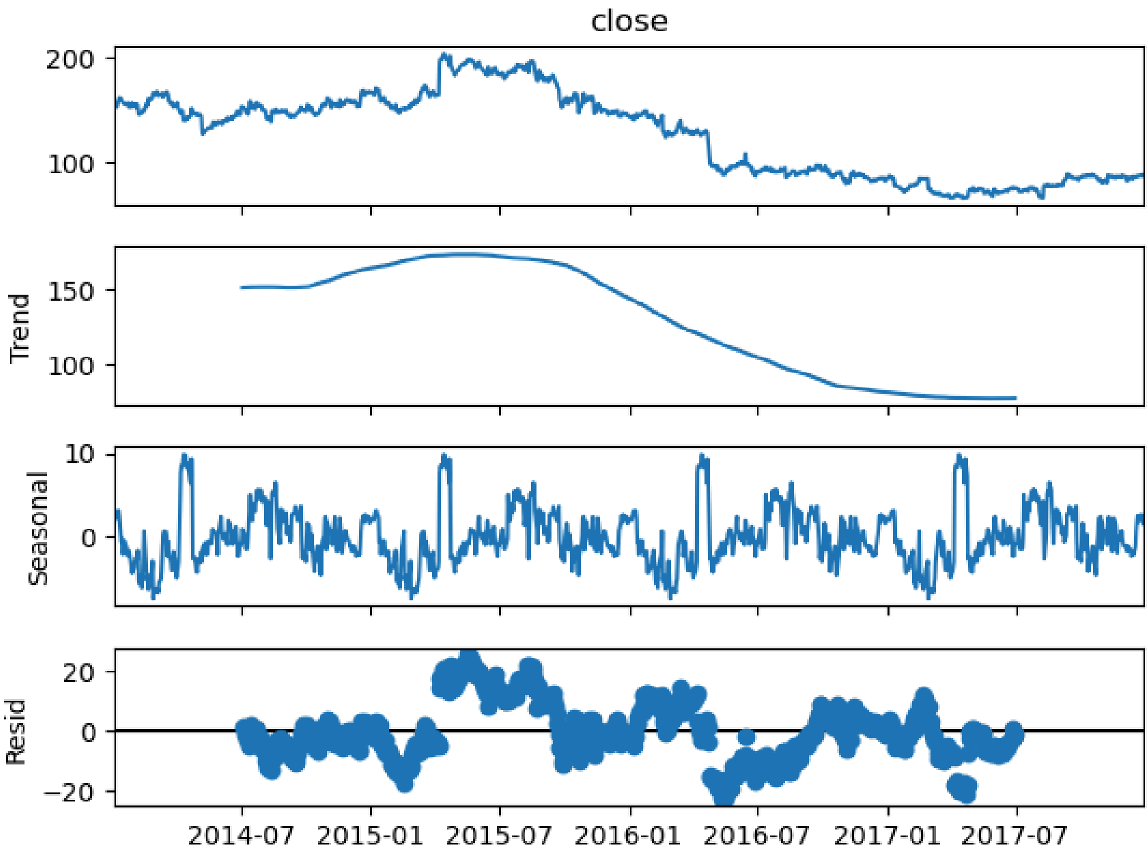
```
prgo = prgo.set_index("date")

prgo.head()
```

Out[25]:

date	symbol	open	high	low	close	volume
2014-01-02	PRGO	152.45	154.52	152.11	153.10	1209456
2014-01-03	PRGO	152.99	154.38	152.61	154.23	712255
2014-01-06	PRGO	154.89	155.27	152.43	152.93	1029774
2014-01-07	PRGO	154.73	155.33	153.71	154.58	1384571
2014-01-08	PRGO	154.61	158.16	154.60	158.03	1720224

In [26]:

```
decomposition = seasonal_decompose(
    prgo["close"],
    model="additive",
    period=252)
decomposition.plot()

plt.show()
```



PRGO started the period with a high price of around $150 and an increase with 2015 recording the peak. From late 2015, the stock entered into a period of prolonged decline. However, towards the end of study period, the stock seemed to strive towards making a recovery. **The Trend** decomposition reveals a stock that transitioned from moderate growth to a long-term decline. The Trend can be grouped into three phases: (1) 2014-2015 where PRGO experienced a remarkable upward trend. (2) Late 2015-2016 where there was a sharp downward movement (3) 2017 where the trend flattens **The Seasonal** decomposition reveals are more irregular compared to NVDA. A recurring pattern was observed , howver, similar to NVDA the magnitude is relatively small compared to the overall decline in stock price. I therefore infer that seasonality is not the primary driver of PRGO's performance **The Residual** component highlights several irregular market movements with breaks at some points. **Overall**, PRGO's persistent downward performance was not primarily influenced by seasonality.

COMPARISON OF NVDA AND PRGO DECOMPOSITION

The decomposition highlights two contrasting stock behaviors. **NVDA** exhibits a strong and accelerating upward trend, indicating sustained growth and increasing investor confidence. In contrast, **PRGO** experiences an early period of growth followed by a prolonged decline, reflecting weakening market performance. While both stocks display recurring seasonal patterns under the assumed annual cycle, these effects are relatively minor compared with their long-term trends. In both cases, the residual components capture irregular price movements caused by unpredictable market events.

MOVING AVERAGE SMOOTHING

1. NVDA MOVING AVERAGE SMOOTHING

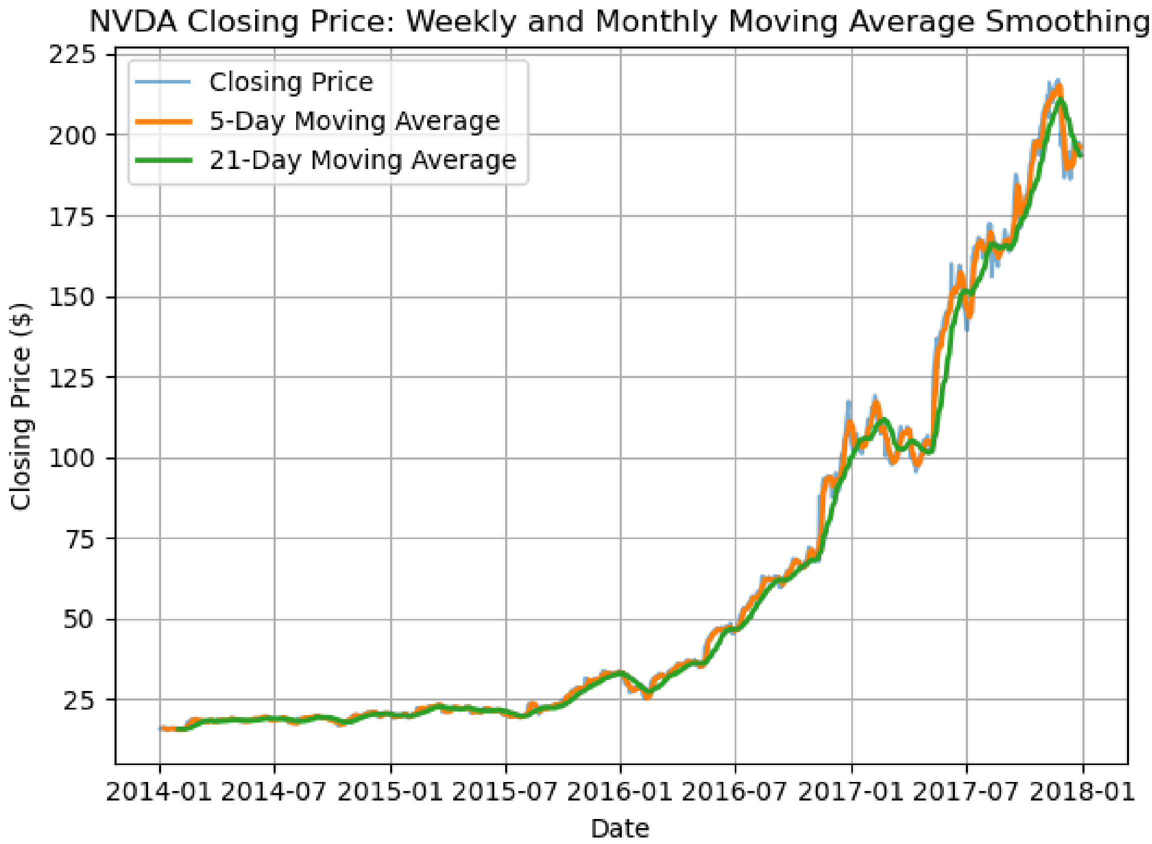
In [27]:
```
nvda["MA_5"] = nvda["close"].rolling(window=5).mean()   # Weekly moving average

nvda["MA_21"] = nvda["close"].rolling(window=21).mean() # Monthly moving average

nvda.head()
```

Out[27]:

date	symbol	open	high	low	close	volume	MA_5	MA_21
2014-01-02	NVDA	15.92	15.9800	15.720	15.86	6502296	NaN	NaN
2014-01-03	NVDA	15.89	15.9201	15.620	15.67	6483288	NaN	NaN
2014-01-06	NVDA	15.83	16.0000	15.680	15.88	10237417	NaN	NaN
2014-01-07	NVDA	16.04	16.2000	15.925	16.14	8332170	NaN	NaN
2014-01-08	NVDA	16.20	16.4400	16.140	16.36	7704980	15.982	NaN

In [28]:
```
plt.plot(nvda.index, nvda["close"], label="Closing Price", alpha=0.6)
plt.plot(nvda.index, nvda["MA_5"], label= "5-Day Moving Average", linewidth=2)
plt.plot(nvda.index, nvda["MA_21"], label= "21-Day Moving Average", linewidth=2)

plt.title("NVDA Closing Price: Weekly and Monthly Moving Average Smoothing")
plt.xlabel("Date")
plt.ylabel("Closing Price ($)")
```

```
plt.legend()
plt.grid(True)
plt.tight_layout()

plt.show()
```



Moving Average Smoothing Interpretation (NVDA)

The moving average analysis highlights the effect of smoothing on NVDA's daily closing prices. The **original closing price** series (Blue line) exhibits frequent short-term fluctuations typical of stock market data. The **5-day** (**weekly**, Orange line) moving average closely follows these movements while reducing minor day-to-day noise, making it suitable to identify short-term trends. In contrast, the **21-day** (**monthly**, green line) moving average produces a smoother curve that emphasizes the medium-term direction of the stock while filtering out much of the daily volatility. Although the monthly moving average responds more slowly to sudden price changes due to its larger averaging window, it provides a clearer representation of the underlying upward trend in NVDA's stock price throughout the study period.

2. MOVING AVERAGE SMOOTHING FOR PRGO

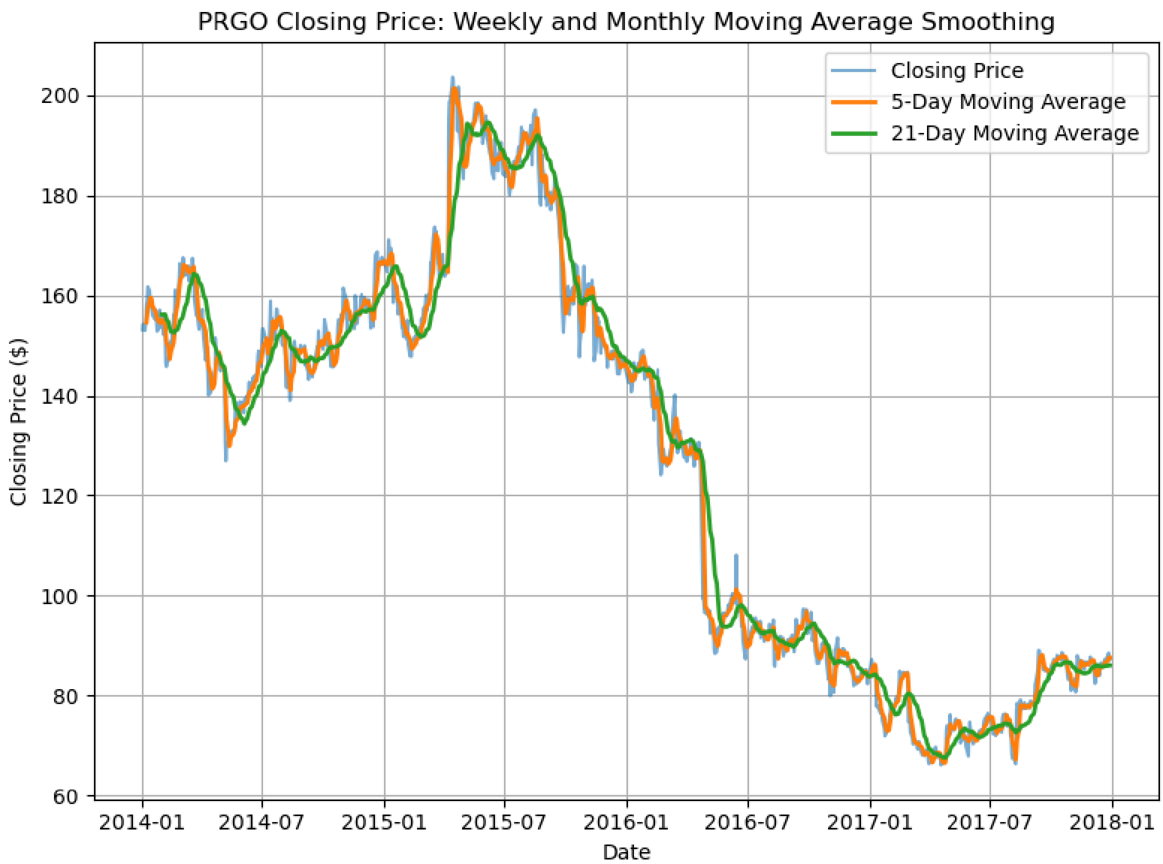
In [29]:
```
prgo["MA_5"] = prgo["close"].rolling(window=5).mean()

prgo["MA_21"] = prgo["close"].rolling(window=21).mean()

prgo.head()
```

Out[29]:

date	symbol	open	high	low	close	volume	MA_5	MA_21
2014-01-02	PRGO	152.45	154.52	152.11	153.10	1209456	NaN	NaN
2014-01-03	PRGO	152.99	154.38	152.61	154.23	712255	NaN	NaN
2014-01-06	PRGO	154.89	155.27	152.43	152.93	1029774	NaN	NaN
2014-01-07	PRGO	154.73	155.33	153.71	154.58	1384571	NaN	NaN
2014-01-08	PRGO	154.61	158.16	154.60	158.03	1720224	154.574	NaN

In [30]:

```
plt.figure(figsize=(8,6))

plt.plot(prgo.index, prgo["close"], label="Closing Price", alpha=0.6)
plt.plot(prgo.index, prgo["MA_5"], label= "5-Day Moving Average", linewidth=2)
plt.plot(prgo.index, prgo["MA_21"], label= "21-Day Moving Average", linewidth=2)

plt.title("PRGO Closing Price: Weekly and Monthly Moving Average Smoothing")
plt.xlabel("Date")
plt.ylabel("Closing Price ($)")
plt.legend()
plt.grid(True)
plt.tight_layout()

plt.show()
```



MOVING AVERAGE SMOOTHING INTERPRETATION- PRGO

The moving average analysis reveals a clear distinction between PRGO's short-term fluctuations and its underlying medium-term trend. The **5-day** moving average closely follows the daily closing price, capturing short-term momentum while reducing some day-to-day noise. The **21-day** moving average produces a smoother series and provides a clearer view of the broader price direction. PRGO experienced considerable price fluctuations during 2014 and 2015, followed by a pronounced downward trend beginning in late 2015. The 21-day moving average clearly captures this sustained decline and the subsequent stabilization and modest recovery during 2017. The increasing lag of the 21-day moving average during sharp price movements demonstrates the trade-off between smoothing and responsiveness.

CONCLUSION

Both Decomposition and Moving Average tell the same story: NVDA and PRGO trended in different closing price trajectoriess during the period of study. **NVDA** exhibited a strong and sustained upward trend, with closing price progressively increasing throughout the latter part of the study period. In contrast, **PRGO** shows considerable short-term fluctuations followed by a prolonged downward trend after its 2015 peak.

Overall, the moving average analysis reinforces the findings from seasonal decomposition: **NVDA experienced sustained growth** during the study period, whereas **PRGO experienced a significant long-term decline** followed by a slight stabilization.